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OF SCIENCE AND TECHNOLOGY

Parameter-Efficient Fine-tuning

ONE LOVE. ONE FUTURE.

Contents

1. Pretraining data
2. PEFT
3. Instruction finetuning



HUST

Pretraining Data

Pretraining data

Benefits of pretraining with more data

Table 11: RoBERTa_{Large} Pretrained with Different Data Setups[10].

Pretrain Setups	Data Size	Batch Size	Steps	MNLI-m (ACC)	SQuAD 1.1 (F1)
Wiki + Books	16GB	8K	100K	89.0	93.6
+Additional Data	160GB	8K	100K	89.3	94.0
+Pretrain More Steps	160GB	8K	300K	90.0	94.4
+Even More Steps	160GB	8K	500K	90.2	94.6

- Same model, same pretraining steps, more data help
- With more data more pretraining steps also help
- Empirical gains more than many “fancier” improvements

Pretraining data

Where to get more pretraining data?

- High quality clean text corpora, such as Wikipedia, news, scientific papers, eventually run out
 - They also have issues, such as copyrights

Web quickly become the only viable option

- The go-to place of general domain human knowledge, in this digital era
- Massive, unlikely grow slower than computing resources
- Publicly available
 - Though with copyright and many other constraints
- Noisy, dirty, and biased

WebText Corpus

Goal: Obtain high quality, large-scale pretraining data from the public web



General Challenges:

- What and which URLs to crawl
- How to extract texts from HTML
- How to select clean texts for pretraining

WebText Corpus



Main questions:

- How to harvest a large number of URLs efficiently
- How to select “high-quality” URLs
 - What makes an URL good?
- How to remove “bad” URLs?
 - Some cases are clear cut: spammy, unsafe, etc.
 - Some are hard to detect or up to debate: certain biases.

WebText Corpus

WebText: The pretraining corpus of GPT-2

- Harvest all outbound links from Reddit
 - Manually mentioned by humans
- Only keep links received ≥ 3 “Karma”
 - A heuristic to filter good URLs
 - Total 45 million URLs deduped to 8 million web pages

How to harvest URLs:

- From Reddit mentions

Definition of good URLs:

- Other Reddit users like it

WebText:

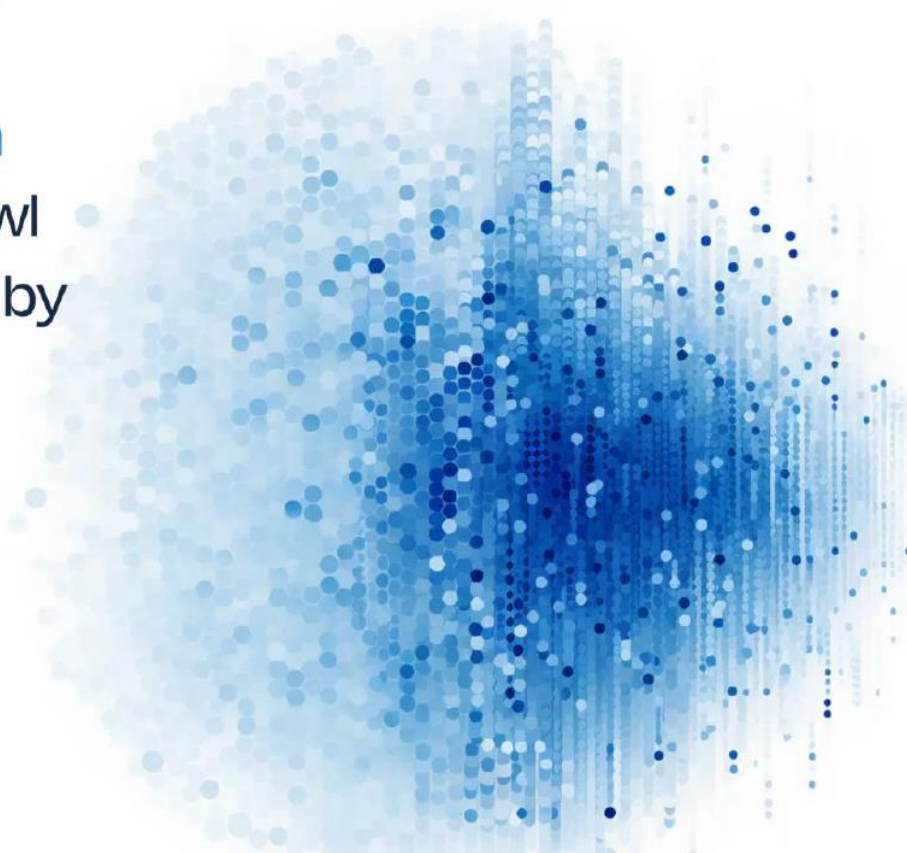
- Pros
 - An easy to harvest a relatively large set of URLs from a common resource
 - Human votes on the URLs
- Bad
 - Reddit starts to forbid the use of its data for pretraining LLMs
 - Limited Scale
 - Reddit is not the cleanest part of the internet

Common Crawl
maintains a **free, open**
repository of web crawl
data that can be used by
anyone.

Common Crawl is a 501(c)(3) non-profit founded in 2007.

We make wholesale extraction, transformation and analysis of
open web data accessible to researchers.

Overview



Common Crawl: commoncrawl.org

- Petabytes of web pages
- Provide open access to large scale web crawls
 - Previously a privilege of search engine companies
- Monthly crawls and dumps
 - Re-crawled web pages and fresh dumps (bi)monthly
 - Recent dumps are ~3 billion pages
 - Date back to past 10 years
 - “220 billion web pages (HTML) captured 2008 – 2021” [10]

Common Crawl Web Exploration:

- Start from a set of seed URLs: popular, high-quality, and trustworthy websites
 - Gov, Edu, etc.
 - Top web domains
- Traverse the web to obtain a candidate set of URLs
 - Around 500 billion links discovered per month crawl
 - About 25+ billion unique ones
- Prioritize a subset (~3 billion) of URLs to crawl and include in the dump

Which URLs to crawl? [11]

- 2008-2012: CC's in-house Page Rank
- 2012-2015: Added ranking and metadata of 22 billion pages donated from web search engine blekko
- 2016-2018: Occasional seed URL donations, ~400 million URLs
- 2016: Alexa and Common Search Rankings
- 2017-Now: CC's in-house web graph based rankings (page rank and centrality)
 - Importance score calculated based on past three-month dumps
 - Steer the crawler for next three month
 - Capped # of URLs per domain

Common Crawl

Top Domains in Common Crawl:

- July 2022 Dump
- Successful Crawls
- Ordered by # of web pages

Favors:

- Sites with more sub-domains
- .edu and .gov
- EU domains

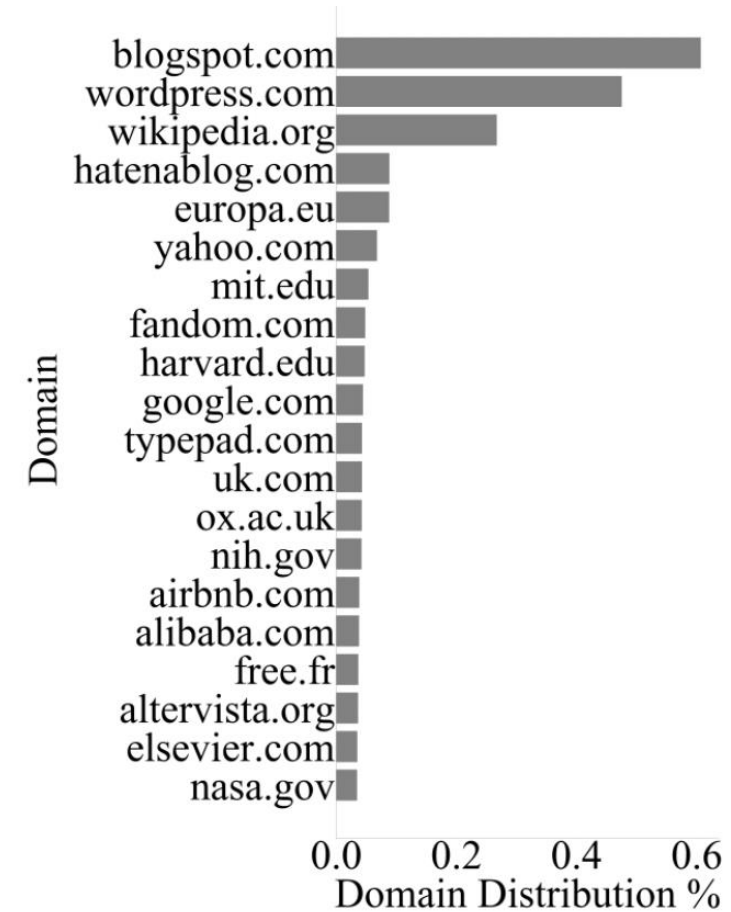


Figure 6: Top 20 domains in Common Crawl July 2022.

Common Crawl

Likely the one and the only public web crawl at this scale

- Still far away from commercial search engines, but the closest we have
- 10 years of crawl dumps enabled various data subsamples
- News, medical, etc.

Accumulation of low resource languages

- One can combine low resources languages from years of dumps to pair with English texts from one month

Pretraining Data Curation from the Web: Scrape Texts



From

```
<header class="post-header"> <h1 class="post-title"> Chenyan Xiong
</h1> <p class="desc">Associate Professor, Language Technologies
Institute, Carnegie Mellon University.</p> </header> <article> <div
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<p>6409 GHC,</p> <p>5000 Forbes Avenue</p> <p>Pittsburgh, PA 15213</p>
</div> </div> <div class="clearfix"> <p>I am an Associate Professor at
Language Technologies Institute (LTI), Carnegie Mellon University
(CMU). My research area is the intersection of information retrieval,
natural language processing, and deep learning. Previously I worked at
Microsoft Research till 2023 Fall, after completing my Ph.D. at LTI,
CMU, in 2018. Before coming to the US, I completed my undergraduate
study at Wuhan University, China, in 2009 and got a master's degree at
the Institute of Software, Chinese Academy of Science, in 2012.</p>
<p>My work mainly appears in IR, NLP, and Machine Learning
conferences. I also host workshops, present tutorials, and serve in
the organization community of venues in these fields. For the most
updated publication list, please refer to my <a
href="https://scholar.google.com/citations?
user=F9BaF8YAAAA&hl=en" rel="external nofollow noopener"
target="_blank">Google Scholar</a>.</p> </div> <div class="social">
<div class="contact-icons"> <a
```

To

Chenyan Xiong.

I am an Associate Professor at Language Technologies Institute (LTI), Carnegie Mellon University (CMU). My research area is the intersection of information retrieval, natural language processing, and deep learning.

Previously I worked at Microsoft Research till 2023 Fall, after completing my Ph.D. at LTI, CMU, in 2018. Before coming to the US, I completed my undergraduate study at Wuhan University, China, in 2009 and got a master's degree at the Institute of Software, Chinese Academy of Science, in 2012.

Scrape Texts from Web Pages: Common Solution

HTML Parsing is a commodity tool for many web related applications

- Many open-source toolkits are available
 - Boilerpipe, selectolax
- Many legacy tools still being used:
 - Common Crawl's official text extraction (WET files) <https://htmlparser.sourceforge.net/> has not updated for 10 years
 - Recent one used to produce refined web: Trafilatura
- Often use a series of rules
 - Identify content nodes and extract the text pieces
 - Remove non-useful HTML codes
 - Glue fragmented text pieces to paragraphs

Pretraining Data Curation from the Web: Clean Texts



Remove noisy, spammy, and fragmented texts

- Non-content texts unlikely to help pretraining

Select higher quality texts from a massive candidate pool

- Given limited pretraining compute budget, pretrain on better texts

Avoid toxic and biased contents

- NSFW contents
- Texts with strong biases

Text Selection for Pretraining



Three common approaches to select clean texts

- Rule-based filtering
- Proximity to high-quality content
- Toxic Classifier

Text Selection for Pretraining: Rule-Based Filtering

Rule-based filtering in Colossal Clean Crawled Corpus (C4, T5 pretraining data)

1. Start from Common Crawl's official extracted texts from HTML
2. Only keep text lines ended with a terminal punctuation mark
3. Discard pages with fewer than 5 sentences
4. Only keep lines with at least 3 words
5. Remove any line with the word "Javascript"
6. Remove any page
 1. with any words in a toxic word dictionary
 2. with the phrase "lorem ipsum"
 3. With "{"
7. De-dup at three-sentence span level

Text Selection for Pretraining: Proximity

Select text sequences “similar” to “good” content

- Good content: often Wikipedia dumps
- Similarity definition:
 - Perplexity on a language model pretrained on good content
 - Classification model trained using good content as positive, rest as negative

Text Selection for Pretraining: Toxic Filter

Remove toxic contents

- By word-based filter (toxic words)
- By toxic classifier
 - Trained using an existing set of toxic texts as filtering target
- Necessary when dealing with open web data

A challenging problem:

- Variant ways of “being toxic”
- Some toxicity is explicit, but some, e.g., biases are implicit
- Definition of toxic varies, often a question for our society

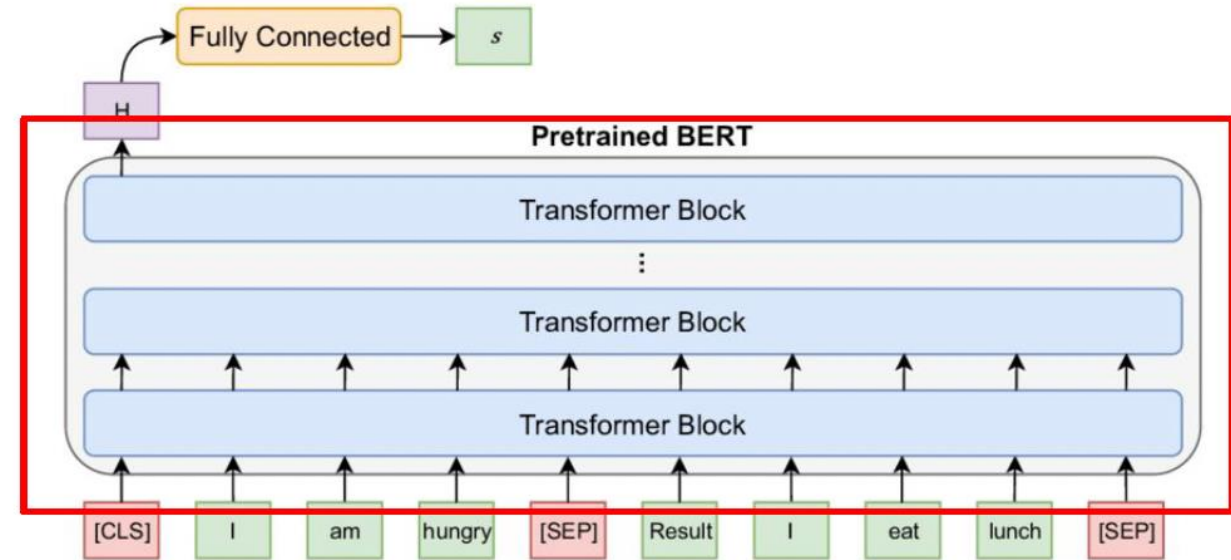
A large graphic on the left side of the slide. It features a dark blue background with a circular pattern of red dots of varying sizes, creating a sense of depth and movement. The word "HUST" is centered within this graphic in a white, bold, sans-serif font.

HUST

PEFT

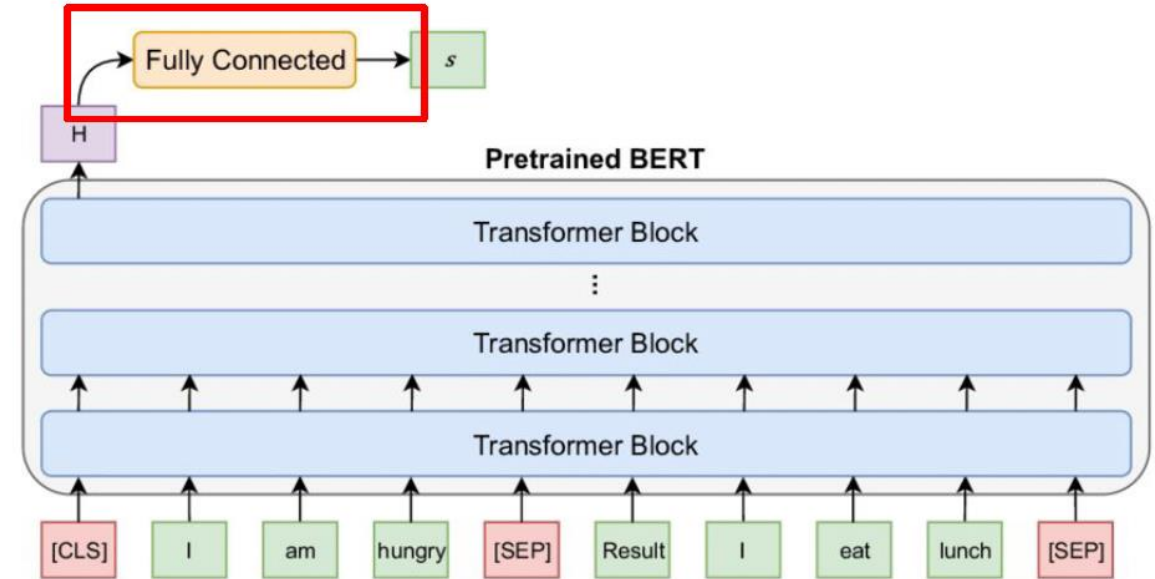
Fine-tuning

- **Pretrain a language model on task**



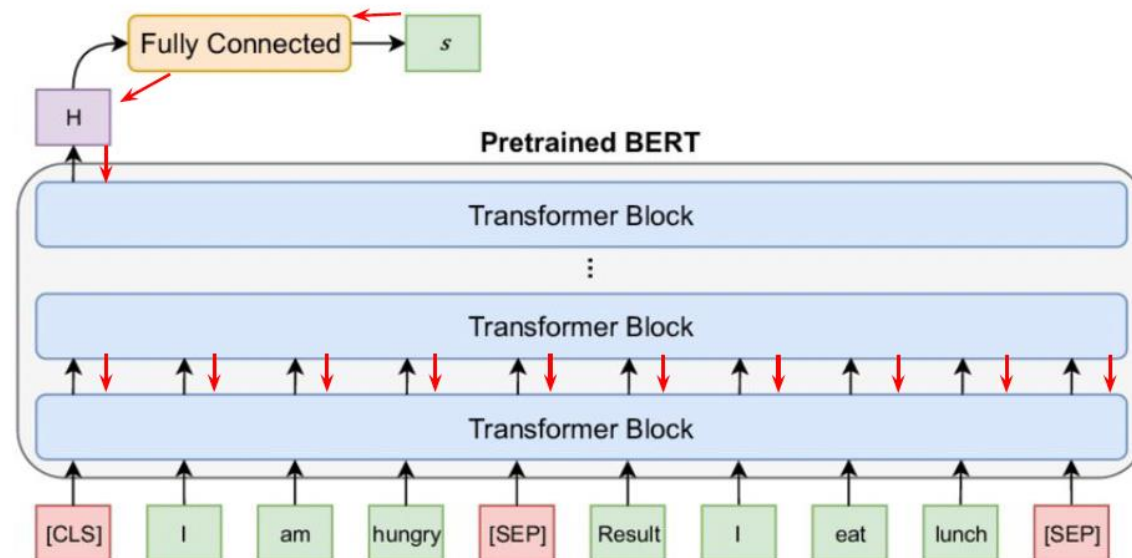
Fine-tuning

- Pretrain a language model on task
- **Attach a small task specific layer**



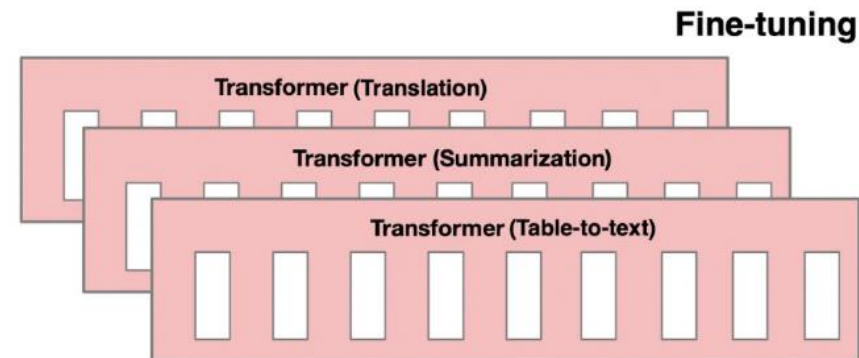
Fine-tuning

- Pretrain a language model on task
- Attach a small task specific layer
- **Fine-tune the weights of full NN by propagating gradients on a downstream task**



Parameter-efficient Fine tuning

- With standard fine-tuning, we need to make a **new copy** of the model for each task.
- In the extreme case of a different model per user, we could never store 1000 different full models.



Parameter-efficient Fine tuning

- With standard fine-tuning, we need to make a **new copy** of the model for each task.
- In the extreme case of a different model per user, we could never store 1000 different full models.
- If we fine tuned a **subset of the parameters** for each task, we could alleviate storage costs. This is **parameter-efficiency**.

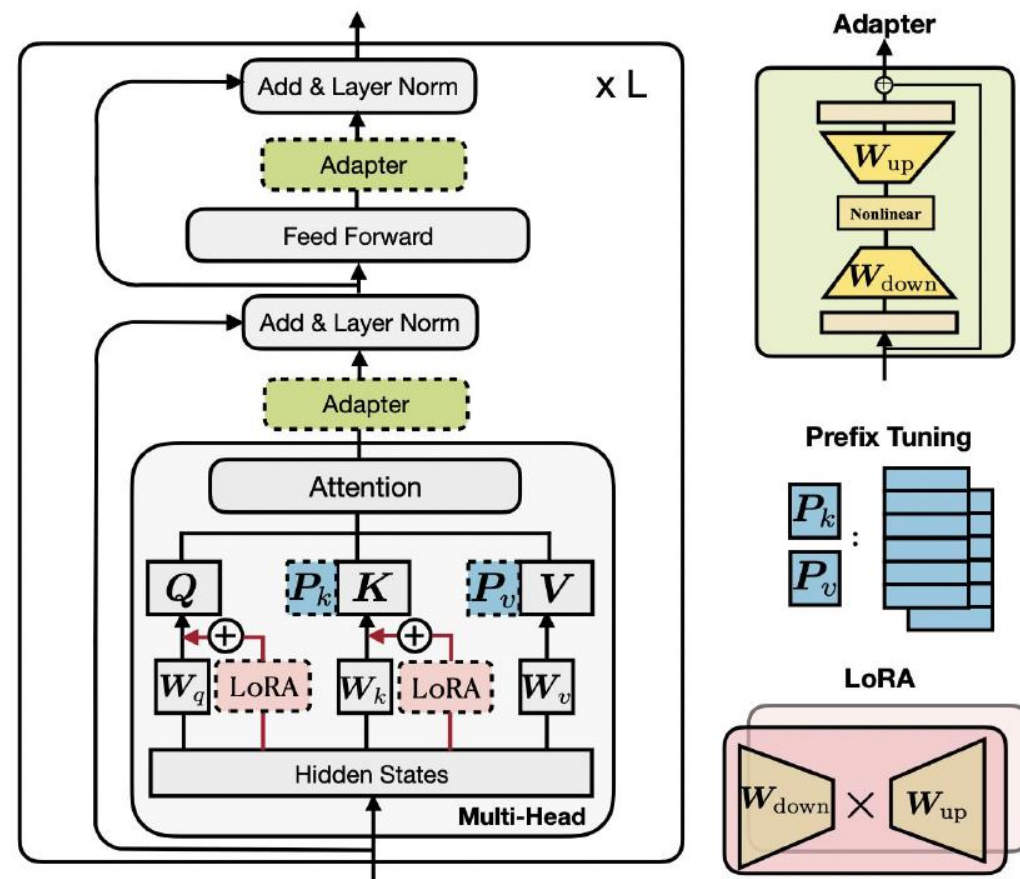
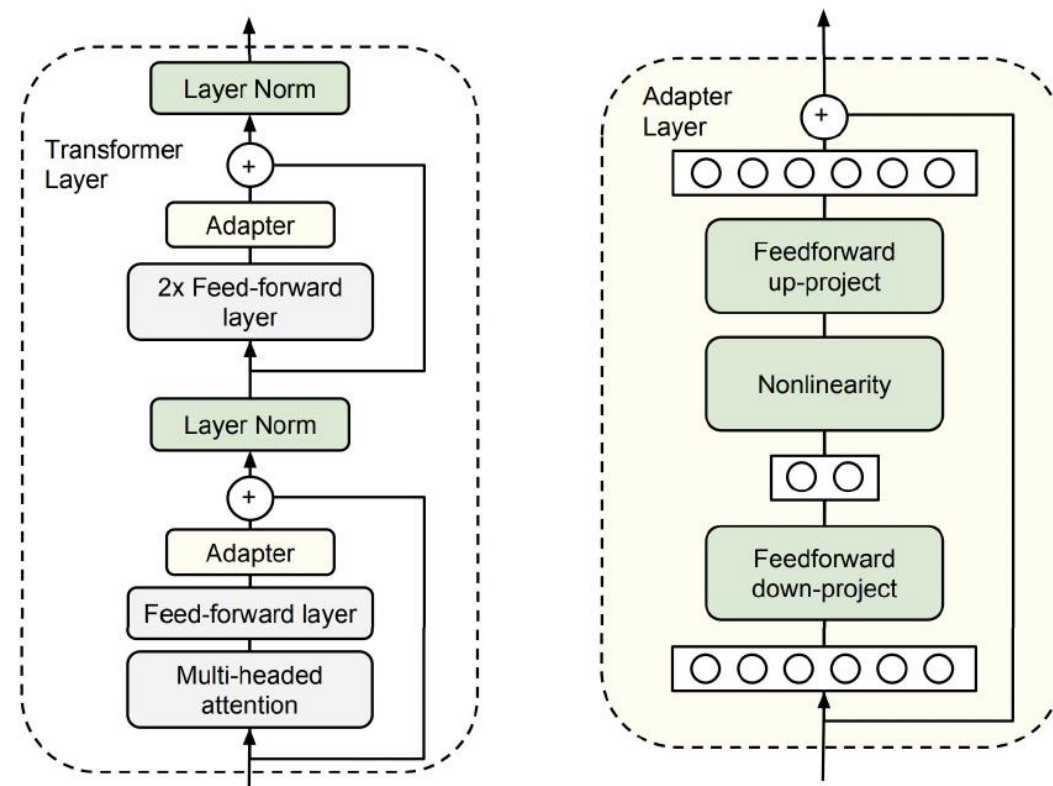


Image: [\(He et al. 2022\)](#)

Adapter Fine Tuning

- They add **adapter layers** in between the transformer layers of a large model.
- During fine-tuning, they **fix the original model parameters** and only tune the adapter layers.
- No need to store a full model for each task, only the adapter params.
- **3.6%** of parameters needed!



(Houlsby et al. 2019)

Prefix-Tuning: Optimizing Continuous Prompts for Generation

Xiang Lisa Li

Stanford University

`xlisali@stanford.edu`

Percy Liang

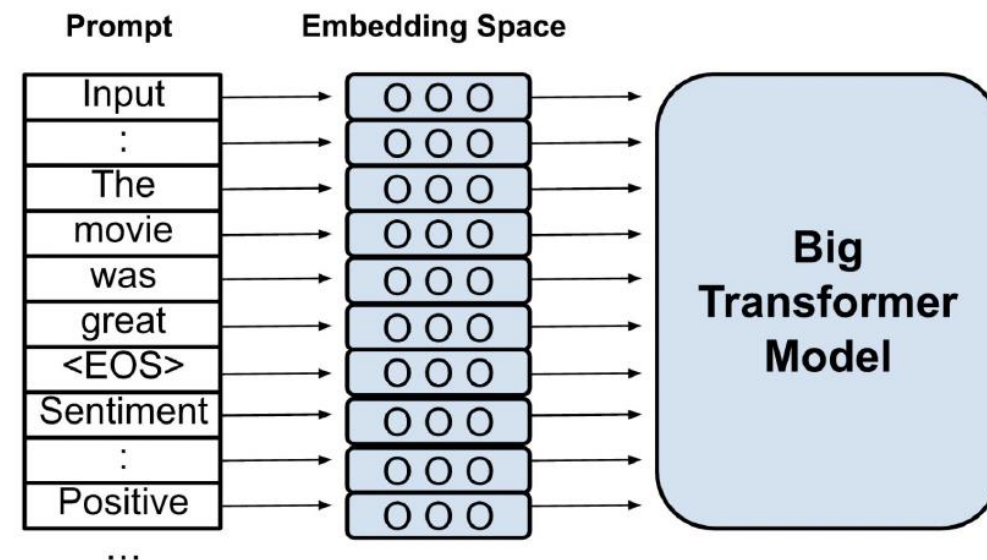
Stanford University

`pliang@cs.stanford.edu`

Published in ACL 2021

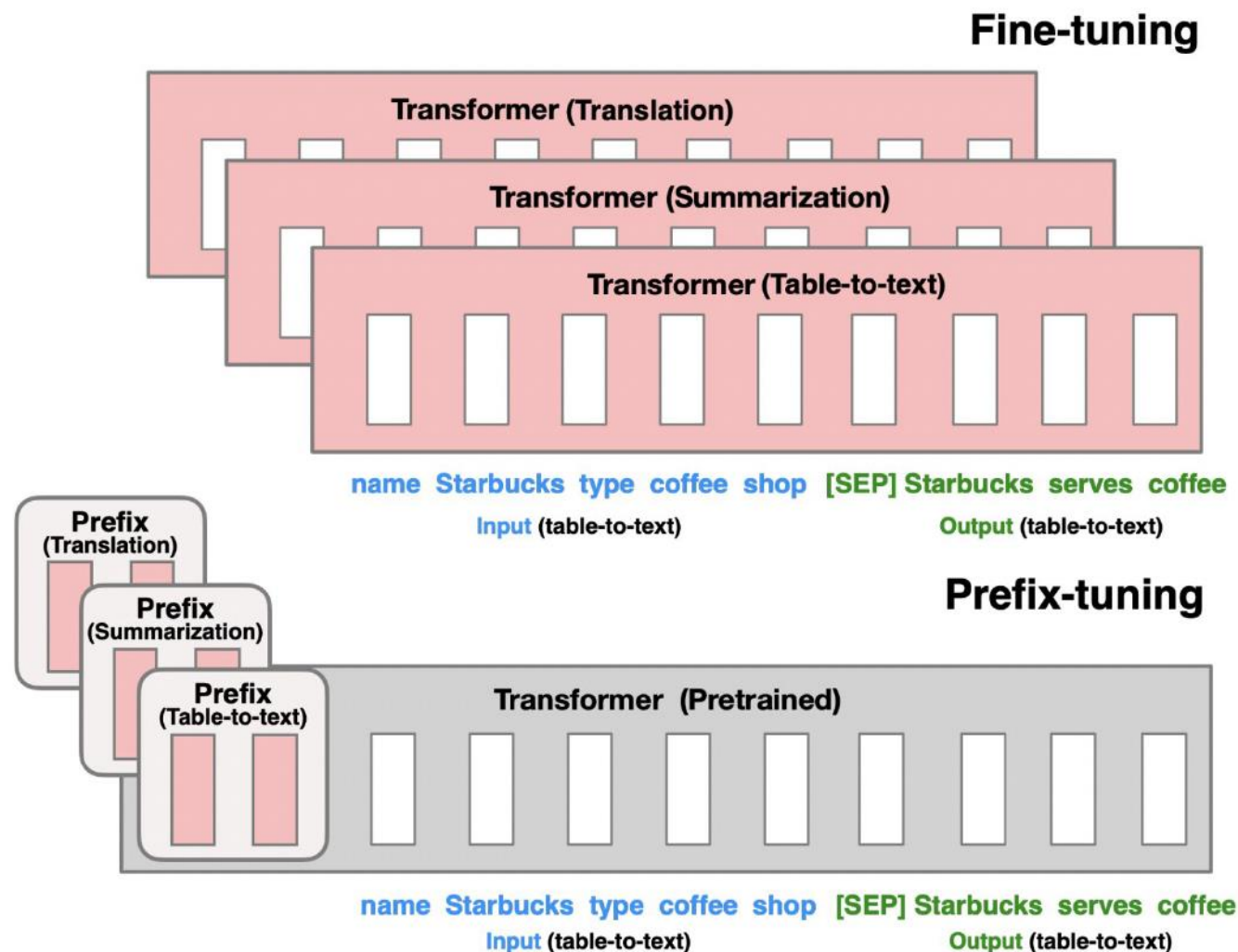
Motivation

- For prompt design ([Brown et al. 2020](#)), the **discrete prompt** is optimized manually.
- Optimization in discrete space is hard! ([Gao et al. 2021](#))
- What if we can optimize the prompt in the **continuous embedding space**?
- This would sacrifice interpretability but would be **easier to optimize**.



Prefix-Tuning

- Rather than designing a prompt manually, we can **learn** an optimal prefix for each task.
- Only **~0.1%** of parameters need to be tuned! (adapter is 3.6%)

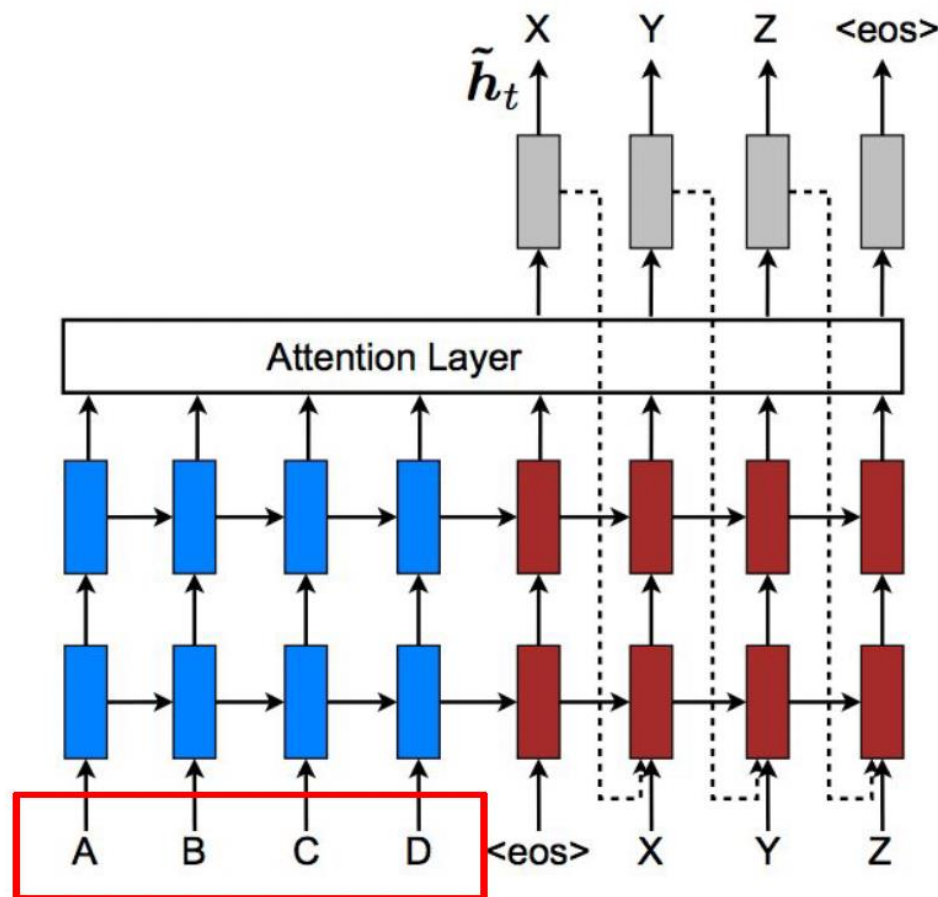


Prefix-Tuning

- Can this be viewed as a continuous relaxation of discrete prompts?

Actually no!

Yay! We just get to tune these now →

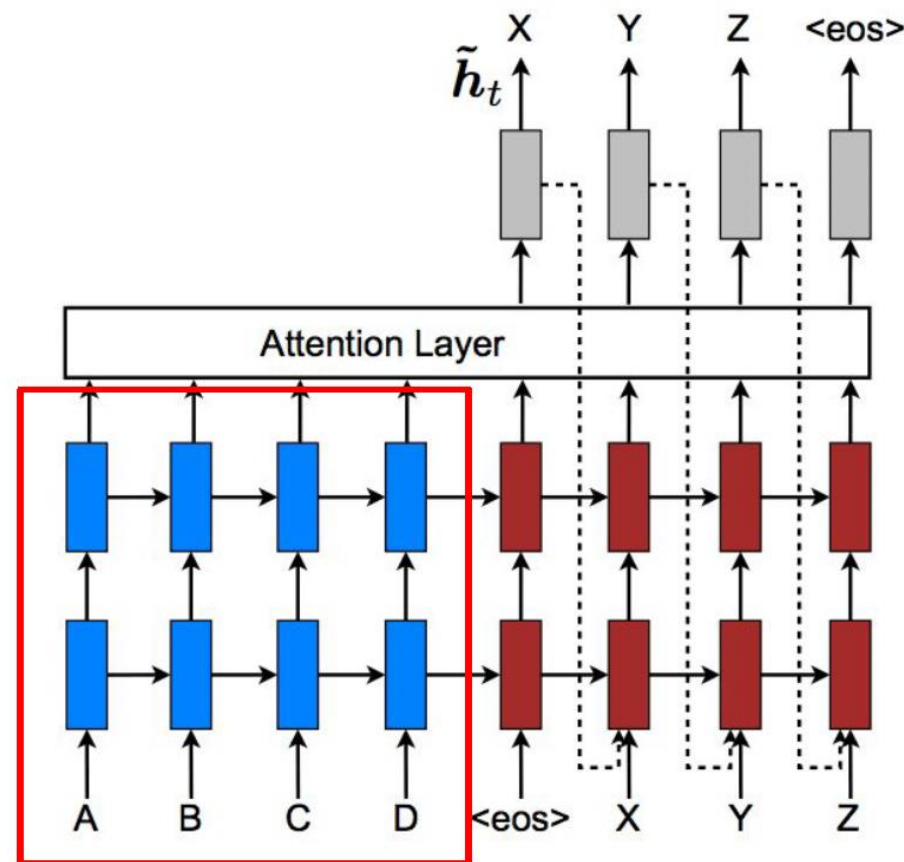


Prefix-Tuning

- Can this be viewed as a continuous relaxation of discrete prompts?

Actually no!

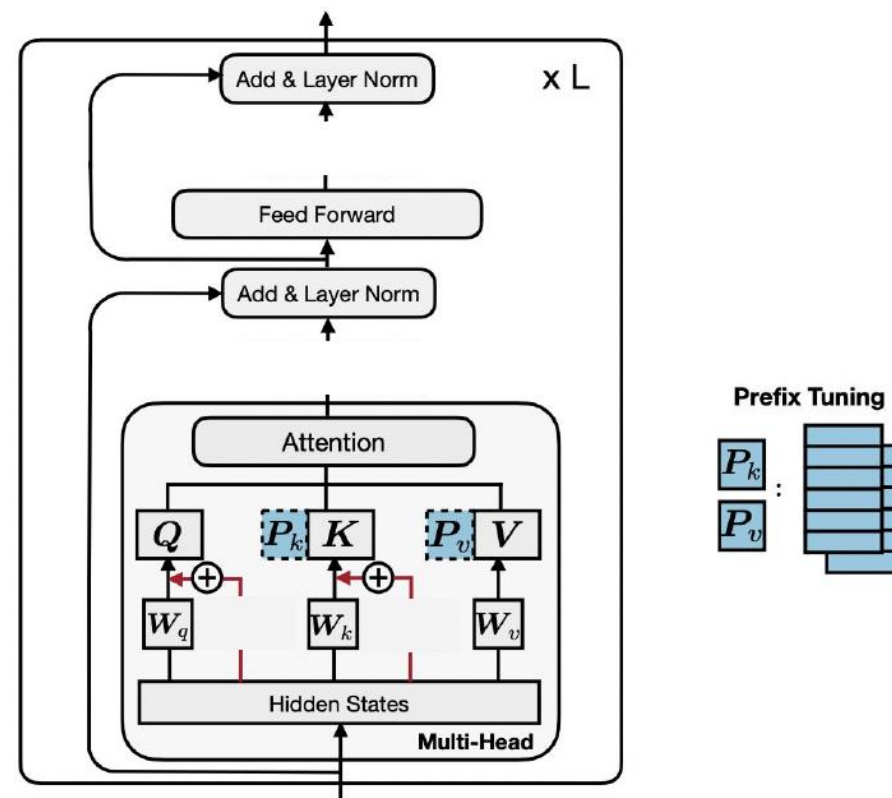
We actually get to tune ALL of these →



Prefix-Tuning

- Can this be viewed as a continuous relaxation of discrete prompts?
- **It's not just the prompts in the prefix that get tuned, it is also the hidden representations of later layer.**

Image: [\(He et al. 2022\)](#)



The Power of Scale for Parameter-Efficient Prompt Tuning

Brian Lester* Rami Al-Rfou Noah Constant

Google Research

`{brianlester, rmyeid, nconstant}@google.com`

Published in EMNLP 2021

Prompt Tuning

- Why do we need **prompt-tuning** if we already have **prefix-tuning**?
 - Prefix-tuning learns a sequence of prefixes that are prepended at **every transformer layer**
 - Prompt-tuning uses a **single** prompt representation that is prepended to the **embedded input**

The “prefix-tuning” paper: *Tuning only the embedding layer is not expressive enough*

- Model scale?
- Prompt length?
- Prompt initialization?
- Pre-training settings?

	E2E				
	BLEU	NIST	MET	ROUGE	CIDEr
PREFIX	69.7	8.81	46.1	71.4	2.49
Embedding-only: EMB-{PrefixLength}					
EMB-1	48.1	3.33	32.1	60.2	1.10
EMB-10	62.2	6.70	38.6	66.4	1.75
EMB-20	61.9	7.11	39.3	65.6	1.85

55

Prompt Tuning

- Prepend **virtual tokens** to input
- Prompt and input representations flow through model like normal
- Learn embeddings of **only these special tokens** via backprop. Keep the rest **fixed**.

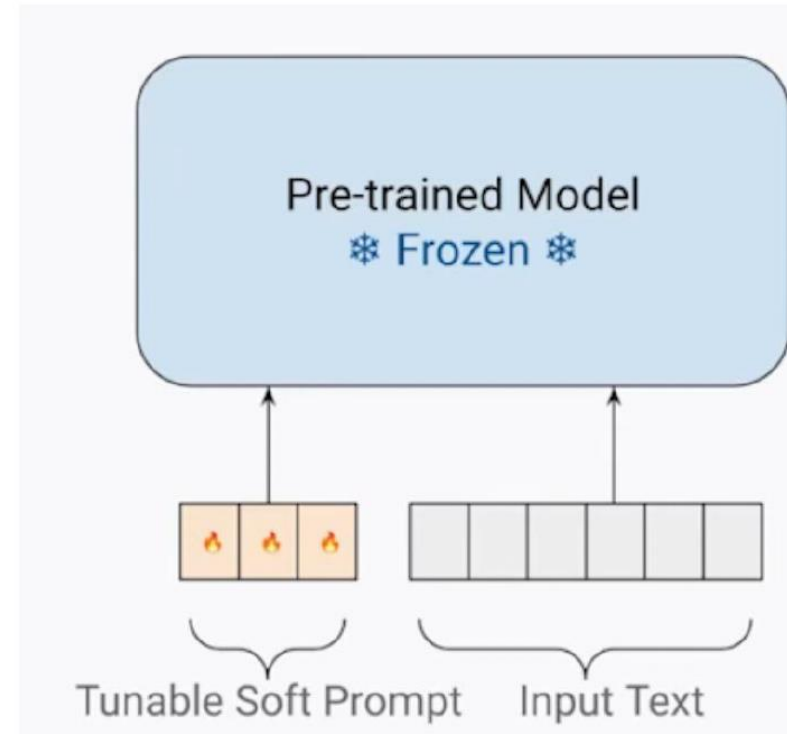
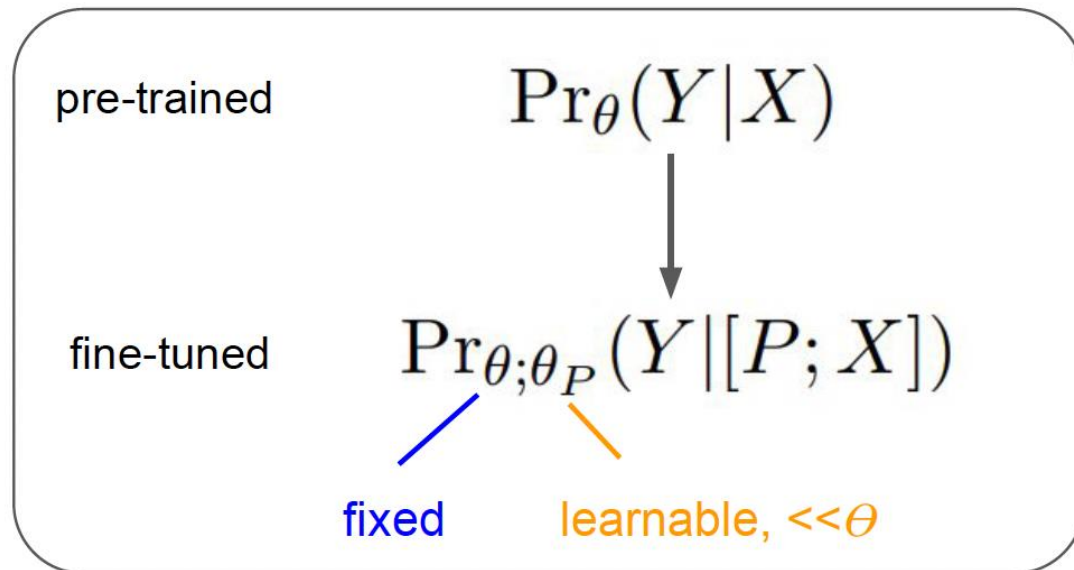


Image Credit: Brian Lester

Low Rank Approximations (LoRA)

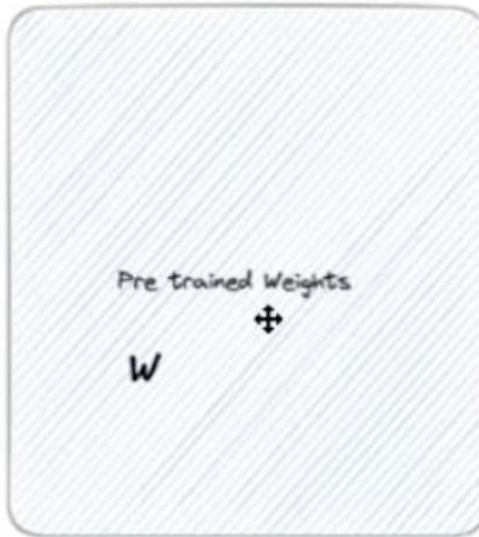
- Improve efficiency by leveraging **low-rank** approximations of the self-attention matrix.
- The key idea is to assume **low-rank structure** in the $N \times N$ matrix.

$$\begin{array}{c} \boxed{M} \\ m \times n \end{array} \approx \begin{array}{c} \boxed{L_k} \\ m \times k \end{array} \times \begin{array}{c} \boxed{R_k^T} \\ k \times n \end{array}$$

Low Rank Approximations (LoRA)

Baseline Full Fine Tuning

h



x



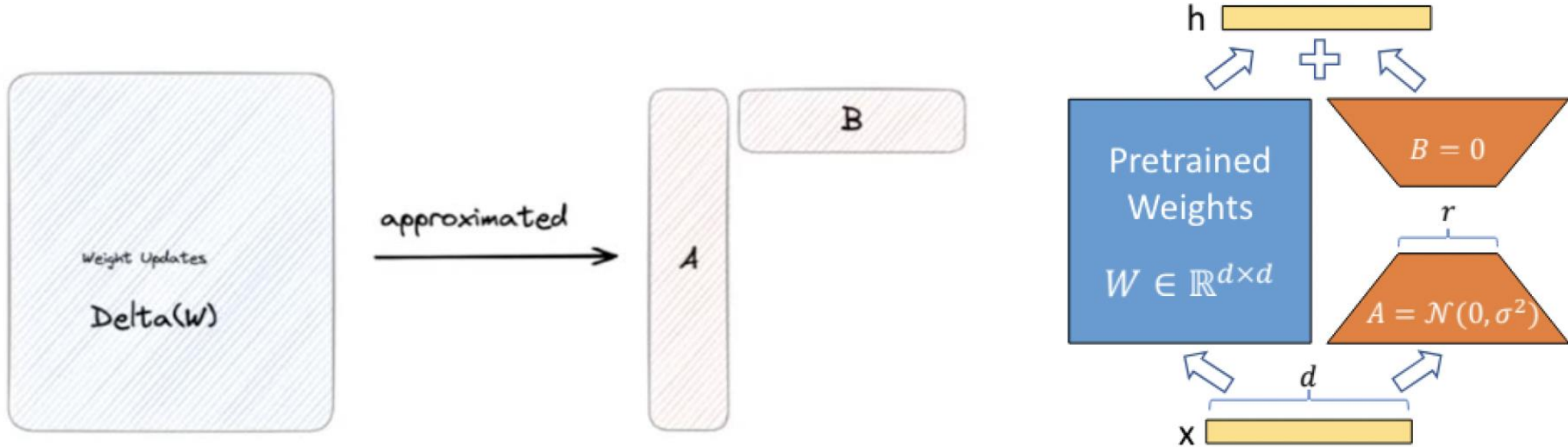
We update weights: $W = W + \Delta(W)$



Problem

$\Delta(W)$ is huge

Low Rank Approximations (LoRA)

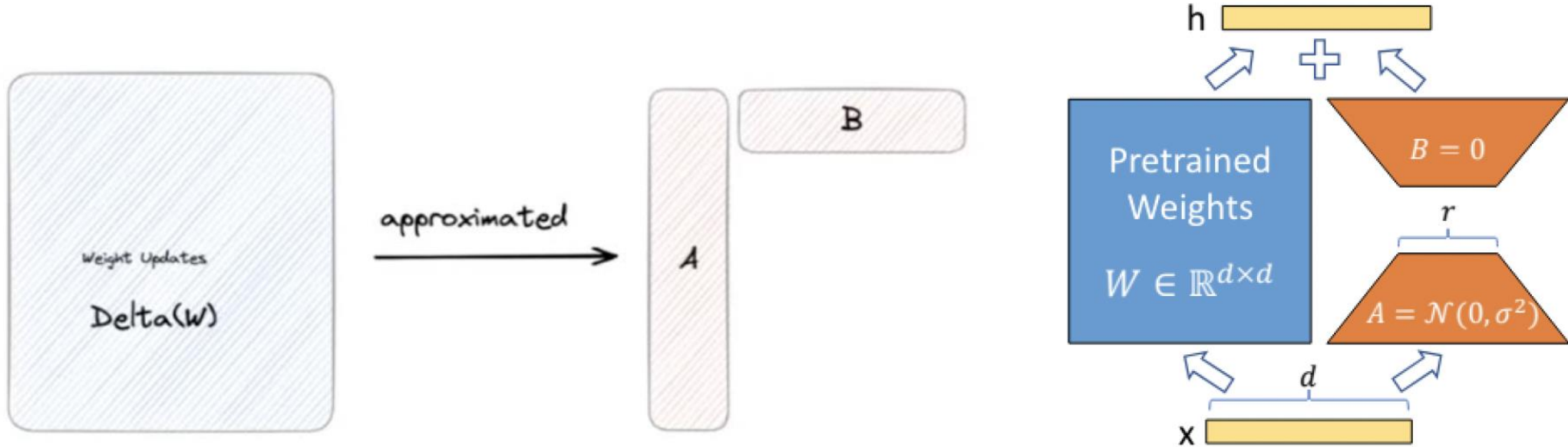


For a pre-trained weights W_0 , we approx $\Delta(w)$ by B and A :

$h = W_0x + \Delta Wx = W_0x + BAx$, where $\bar{B} \in \mathbb{R}^{d \times r}$, $\bar{A} \in \mathbb{R}^{r \times k}$, and the rank $r \ll \min(d, k)$

During Training, we only compute gradient w.r.t. $\Delta(W)$

Low Rank Approximations (LoRA)



For a pre-trained weights W_0 , we approx $\Delta(w)$ by B and A :

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HUST

Instruction Fine-tuning

Motivation of instruction finetuning

Language modeling $\neq$ assisting users

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION

GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

Language models are not *aligned* with user intent.
Do **completion** instead of instruction following

Motivation of instruction finetuning

Instruction: Given an address and city, come up with the zip code.

Input:

Address: 123 Main Street, City: San Francisco

Output: 94105



Instruction: I am looking for a job and I need to fill out an application form. Can you please help me complete it?

Input:

Application Form:

Name: _____ Age: _____ Sex: _____

Phone Number: _____ Email Address: _____

Education: _____ ...

Output:

Name: John Doe Age: 25 Sex: Male

Phone Number: ...



Instruction: How to write a code for converting degrees fahrenheit to celsius.

Input: Null

Output:

```
def convert_fahrenheit_to_celsius(fahr):  
    celsius = (fahr - 32) * 5 / 9  
    return celsius
```



Instruction: Write a letter from the perspective of a cat.

Input: Null

Output:

Dear [Owner], I am writing to you today because I have a problem. You see, I've been living with you for the past [Number of Years] and I love it here. The food is great, the bed is comfy, and there are plenty of toys to play with. But lately, I've noticed that you're not as affectionate towards me as you used to be. In fact, sometimes when I try to cuddle up next to you on the couch, you push me away ...



```
{"id": "seed_task_8", "name": "english_haiku_generation",  
"instruction": "请以下面词语为主题写一首诗", "instances": [{"input": "夏天", "output": "不但春妍夏亦佳，随缘花草是生涯。\\n鹿葱解插纤长柄，金凤仍开最小花。"}], "is_classification": false}
```

```
{"id": "seed_task_38", "name": "synonym_generation", "instruction":  
"给出下面词语的同义词", "instances": [{"input": "惊人地", "output": "令人惊奇地，令人惊讶地，意外地，难以置信地，不可思议地"}],  
"is_classification": false}
```

```
{"id": "seed_task_44", "name": "add_to_the_list", "instruction": "根据  
【】内的提示，续写下面的内容", "instances": [{"input": "我认为在夏天，狗狗可能喜欢吃西瓜、冰冻花生酱、【它们平时吃的食物】",  
"output": "水管里的水、冰块、冷肉"}], "is_classification": false}
```

Wang, Y., Kordi, Y., Mishra, S., Liu, A., Smith, N.A., Khashabi, D. and Hajishirzi, H., 2022. Self-instruct: Aligning language model with self generated instructions. *arXiv preprint arXiv:2212.10560*.

Superficial Alignment Hypothesis

Alignment is to learn the **response format or the interaction style** ! (Task Recognition)

It is enough to use **1030 examples** for Superficial Alignment [1]

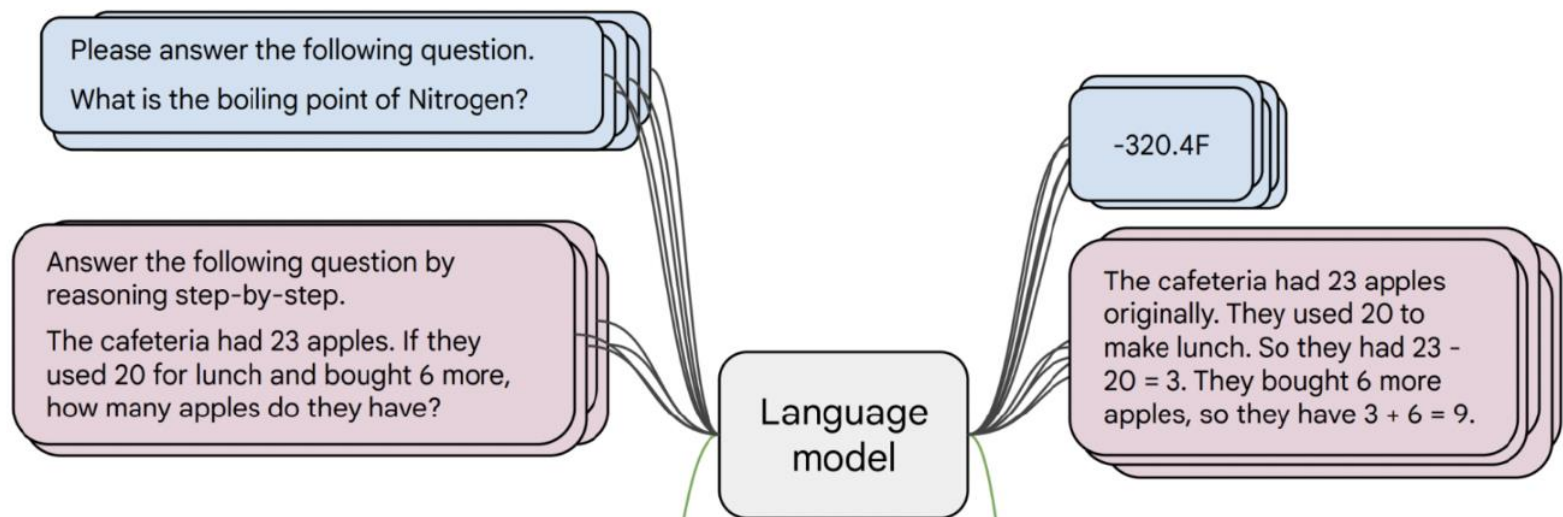
- 1000 examples for instruction following
- 30 examples for conversation

Less is more?

[1] Chunting Zhou, Pengfei Liu, Puxin Xu, Srini Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, Omer Levy. LIMA: Less Is More for Alignment. <https://arxiv.org/abs/2305.11206>
Chen, Hao, et al. "Maybe Only 0.5% Data is Needed: A Preliminary Exploration of Low Training Data Instruction Tuning." arXiv preprint arXiv:2305.09246 (2023).

Task generalization: FLAN T5

- **Collect examples** of (instruction, output) pairs across many tasks and finetune an LM



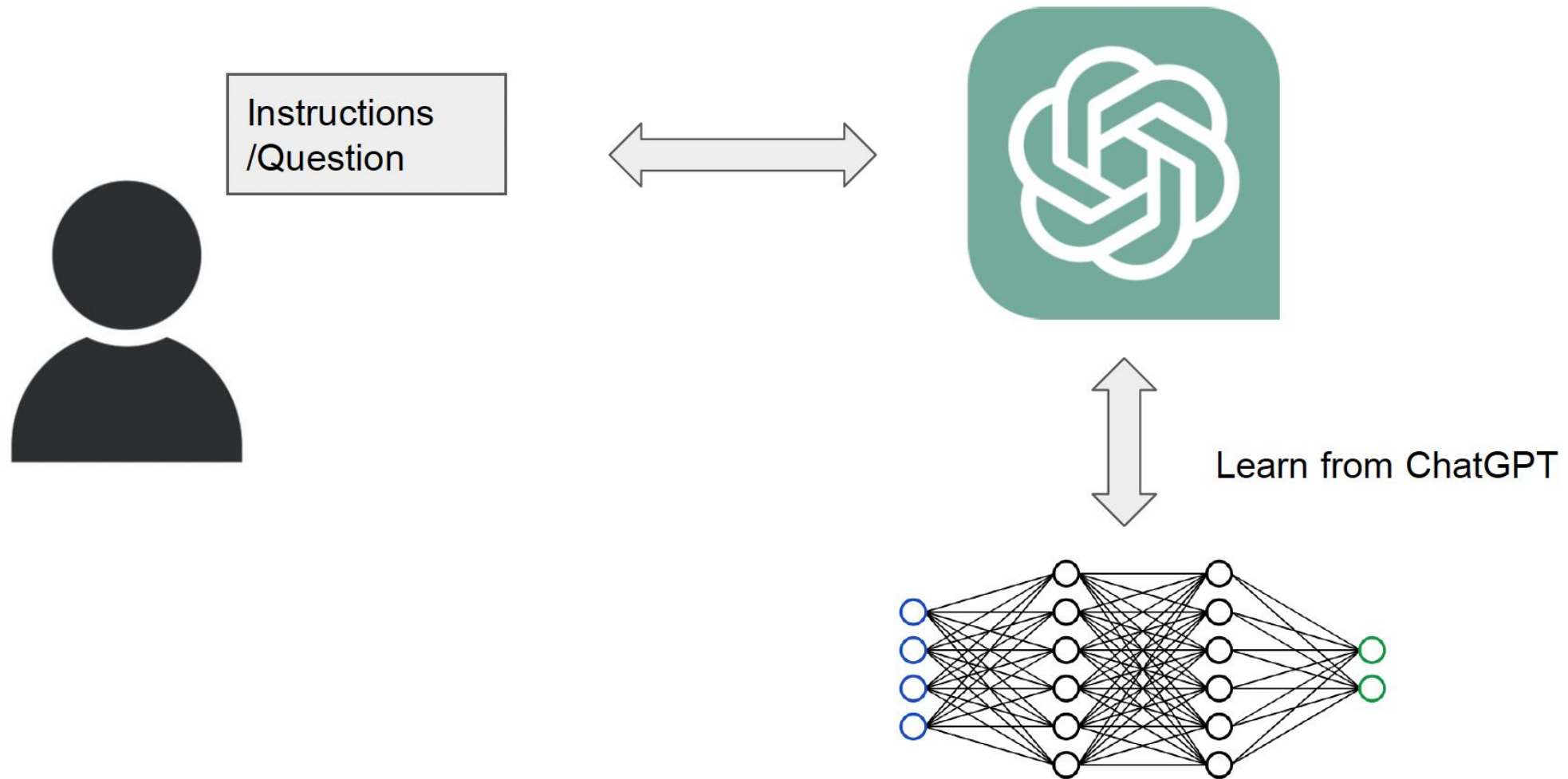
- **Evaluate on unseen tasks**



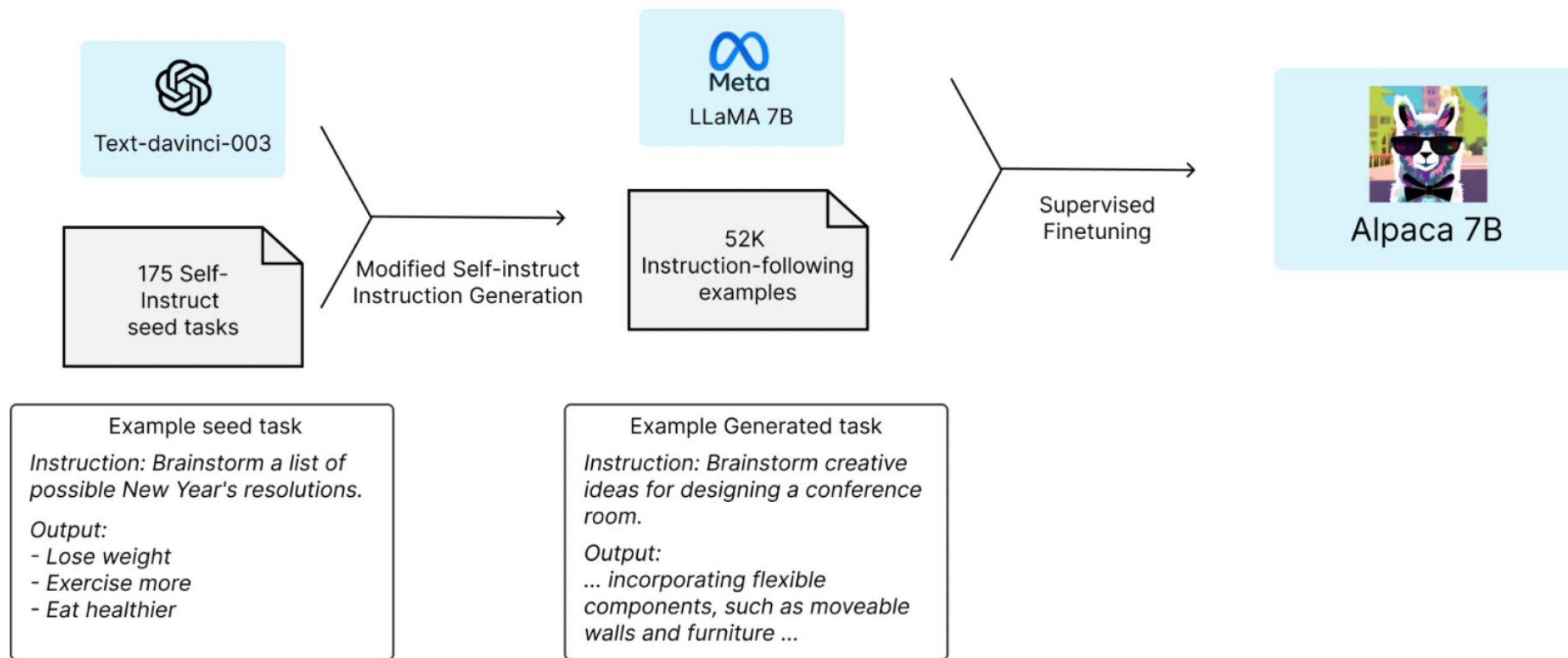
More **diverse**, the better

Wei, Jason, et al. "Finetuned language models are zero-shot learners." arXiv preprint arXiv:2109.01652 (2021).

Shortcut: Distillation from ChatGPT



Tip 1: Self-instruct for data augmentation



Tip 2: training on output only

Single-turn:

System_Prompt + <User>: [User_Input] +<System>: [Response]</s>

Loss

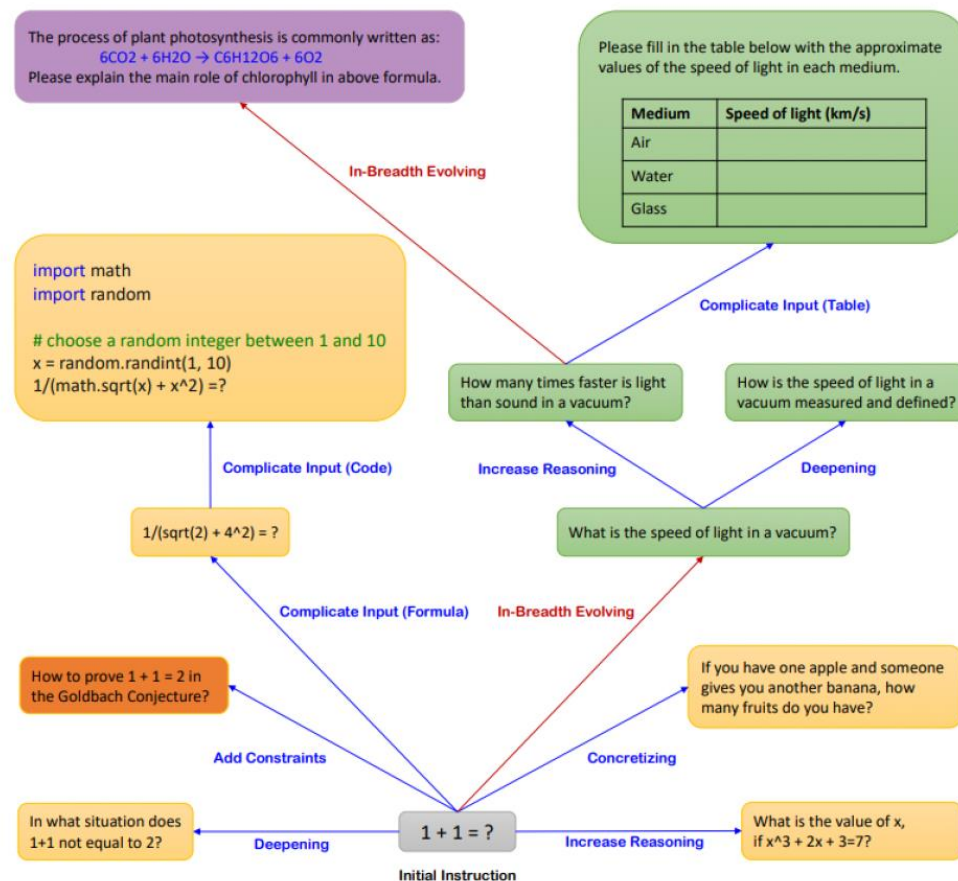
Multi-turn:

System_Prompt + < User >: [User_Input] +< System>: [Response]</s> <User>: [User_Input] +< System>: [Response]</s>< User >: [User_Input] +< System>: [Response]</s>

Loss

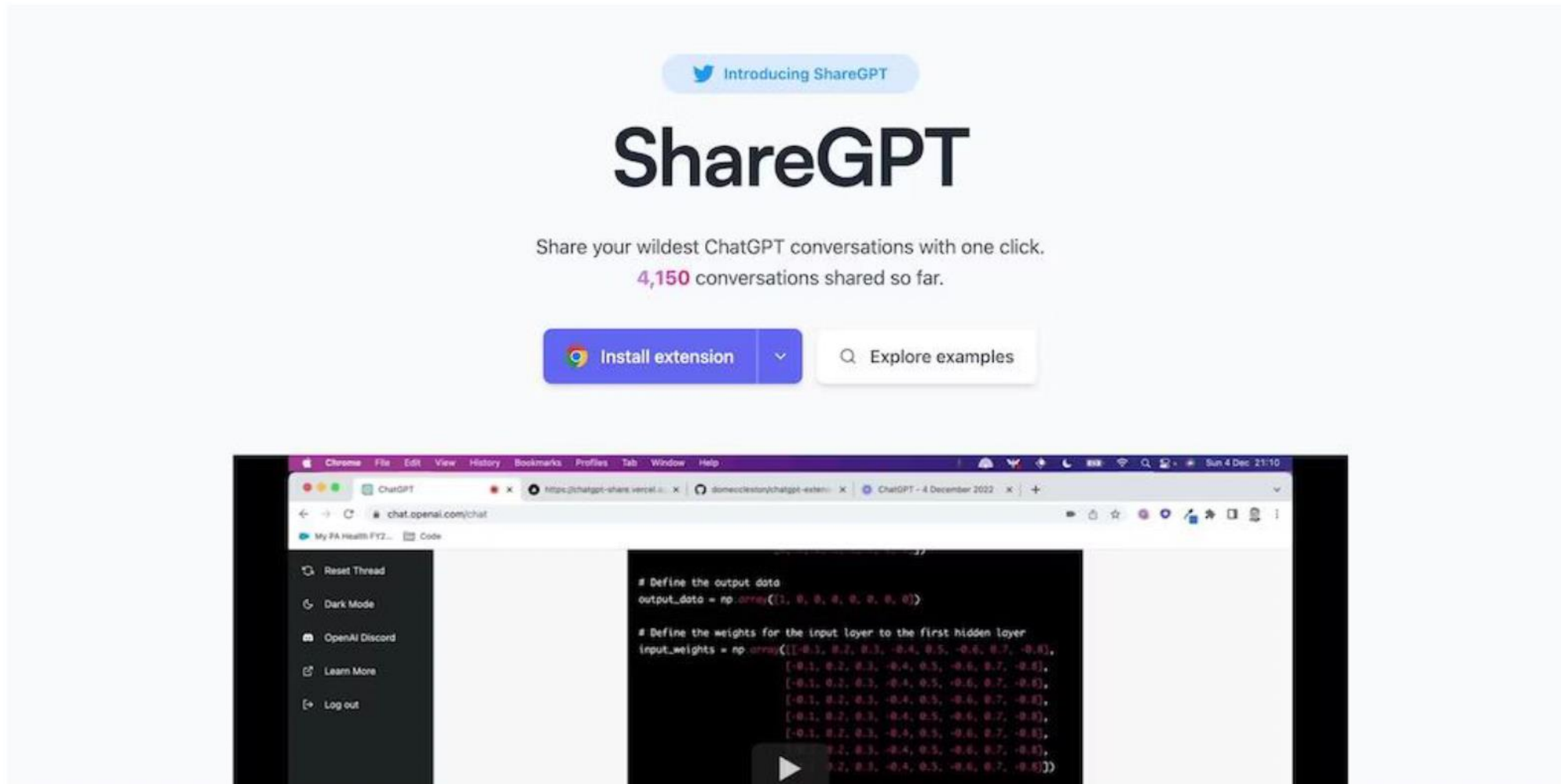
Tip 3: Use complex instructions

WizardLM: Empowering Large Language Models to Follow **Complex** Instructions



Xu, C., Sun, Q., Zheng, K., Geng, X., Zhao, P., Feng, J., Tao, C. and Jiang, D., 2023. Wizardlm: Empowering large language models to follow complex instructions. *arXiv preprint arXiv:2304.12244*.

Instruction tuning in multiturn (Conversation)



Why ShareGPT is great ?

What is the meaning of life? Explain it in 5 paragraphs.

The meaning of life is a philosophical question that has been debated throughout history. It is a complex and multifaceted topic, and different people may have different answers depending on their beliefs, values, and experiences. In general, though, the meaning of life is thought to be the reason for which we exist, the purpose that gives our lives significance and direction.

One possible answer to the question of the meaning of life is that it is to seek happiness and fulfillment. This is a common theme in many philosophical and religious traditions, which often emphasize the importance of living a good and virtuous life in order to achieve happiness and fulfillment. For some, this may mean striving for personal growth and self-improvement, while for others it may involve pursuing spiritual enlightenment or a connection with a higher power.

Another possible answer to the question of the meaning of life is that it is to contribute to the greater good. This approach emphasizes the importance of serving others and making the world a better place, whether through acts of kindness, charitable work, or other forms of service. For those who believe in this approach, the meaning of life is not just about individual happiness, but also about making a positive impact on the world and the lives of others.

A third possible answer to the question of the meaning of life is that it is to discover and pursue one's passions and interests. This approach emphasizes the importance of living a life that is authentic and true to oneself, and of pursuing one's passions and interests with enthusiasm and dedication. For those who believe in this approach, the meaning of life is not just about achieving some external goal, but about living a life that is rich, fulfilling, and true to who we are.

In conclusion, the meaning of life is a deeply personal and subjective question, and different people may have different answers. Some may believe that the meaning of life is to seek happiness and fulfillment, while others may believe that it is to contribute to the greater good or to pursue one's passions and interests. Ultimately, the meaning of life is a mystery that may never be fully understood, but that we can strive to uncover and understand through our experiences, beliefs, and values.

Shouldn't the answer be 42?

The answer "42" is a reference to a famous line in the book "The Hitchhiker's Guide to the Galaxy" by Douglas Adams. In the book, the character Deep Thought is asked to find the answer to the ultimate question of life, the universe, and everything. After thinking for a very long time, Deep Thought reveals that the answer is "42." However, the characters in the book then realize that they don't actually know what the question is, so the answer is ultimately meaningless.

In other words, the answer "42" is a humorous and satirical take on the idea that there may be a single, definitive answer to the ultimate question of life, the universe, and everything. It suggests that the search for such an answer may be futile, and that the meaning of life is ultimately unknowable. Therefore, while "42" is a popular meme, it is not necessarily a meaningful answer.

ShareGPT
Powered by Vercel

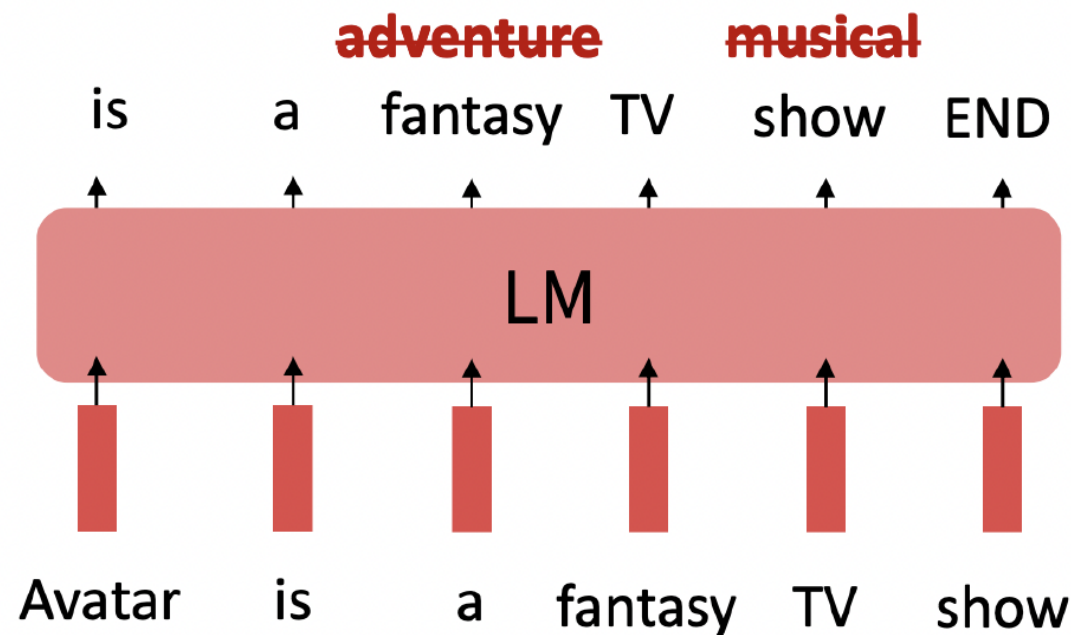
Copy 125 41.6K

- User questions/instructions are **real**!
- Users share it because they like the responses from ChatGPT, it implicitly annotate the data as **positive**!

However, ShareGPT is not able to be downloaded since April; we have limited ShareGPT data, which is not scalable.

Limitations of Instruction Finetuning

- **Expensive** to collect groundtruth data for so many tasks.
- Tasks like open-ended creative generation **have no right answer**.
 - Write me a story about a dog and her pet grasshopper.
- Language modeling **penalizes** all token-level mistakes **equally**, but some errors are worse than others.
- Mismatch between LM objective and human preferences



Can we explicitly attempt to satisfy human preferences?

References

1. COS 597G (Fall 2022): Understanding Large Language Models
<https://www.cs.princeton.edu/courses/archive/fall22/cos597G/>
2. CSC 6203 Large Language Models: <https://llm-course.github.io/>



HUST

Xin trân trọng cảm ơn!